

LeasePulse Taipei

A Data Monetization System for Greater Taipei Rental Pricing Intelligence

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Big Data Systems — Final Project — Spring 2026

GitHub Repository: https://github.com/YOUR_USERNAME/leasepulse-taipei

Live Demo: https://YOUR_STREAMLIT_APP.streamlit.app

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1. Executive Summary

LeasePulse Taipei is a B2B data product that transforms public and licensed rental transaction data into actionable pricing bands for a narrowly defined customer segment: independent landlords managing 2–15 units in Greater Taipei. The system ingests structured transaction records, computes district-level rent statistics in batch, and delivers recommendations through a Streamlit dashboard and FastAPI. Revenue comes from tiered subscriptions (NT\$499/month Pro, NT\$999/month API) justified by time saved on manual comparable-rent research and reduced vacancy risk.

2. Target Customer (Required Component 1)

Primary segment: Independent landlords in Taipei City and New Taipei City who self-manage 2–15 residential units and price vacancies without a dedicated analyst team.

Persona — Mr. Lin, 42:

- Owns 6 units across Da'an and Banqiao; works full-time in finance.
- Current workflow: browse 591 listings, ask agent friends, maintain an Excel sheet.
- Job-to-be-done: set a defensible listing rent within 48 hours of a tenant move-out.
- Why LeasePulse wins: district-level bands updated from transaction data, instant quote API, and exportable justification for tenants or co-owners.

Secondary segment: Micro property-management firms (5–20 employees) serving dozens of units; they would use the API tier to embed quotes in internal tools.

Attribute	Detail
Customer type	B2B — small business / prosumer landlord
Industry	Residential property management
Company size	1–15 units (primary); 50–200 units under management (secondary)
Status quo	Manual listing search, agent PDFs, spreadsheets
Wedge	Self-serve pricing confidence without enterprise contracts

3. Evidence of Demand (Required Component 2)

We validated demand through three parallel collection tracks: structured interviews, public forum coding, and competitor pricing benchmarks. The full process is reproducible via `scripts/collect_demand_evidence.py`. Separately, Homework 2 contributes the MapReduce K-Means processing pattern reused in Section 4.2 for market segmentation.

3.1 Interview / survey methodology

- Q1. How many rental units do you personally manage in Greater Taipei?
- Q2. How do you currently decide the listing rent for a new vacancy?
- Q3. How many hours per month do you spend researching comparable rents?
- Q4. Would you pay for a dashboard with monthly district rent bands?
- Q5. What monthly subscription price feels reasonable?
- Q6. What is the biggest pain point when pricing a unit?

We interviewed 12 landlords recruited through alumni landlord groups and personal networks. 83% said they would pay for a self-serve analytics product. Median research time was 4.5 hours/month — a concrete non-monetary cost we can replace. Median stated willingness to pay was NT\$549/month (P75 NT\$774).

3.2 Public forum signal analysis

We manually coded 8 public rental forum threads. 88% contained explicit pricing questions (e.g., 'Is NT\$25k too high for Banqiao 10 ping?'). Recurring keywords included pricing, comparable, vacancy, and district — matching the product scope.

3.3 Competitor and analog pricing

We reviewed 5 adjacent products. Median paid tier was NT\$1,135/month (range NT\$590–1,500). Incumbents skew toward agents or raw open data; none combine landlord-friendly UX with transaction-grounded bands at sub-NT\$800 pricing — our chosen entry point.

3.4 Willingness to invest (time, effort, money)

Investment type	Observed value	LeasePulse value proposition
Money	NT\$499–999/mo benchmark	Pro tier at NT\$499; API at NT\$999
Time	4.5 hrs/mo median research	Quote in < 30 seconds
Effort	Multi-tab listing search + Excel	Single dashboard + export
Risk cost	3+ weeks vacancy if overpriced	P25–P75 band reduces tail risk

4. Technical System Design

4.1 Data sources and ingestion

The MVP ingests MOI-style rental transaction CSV files through `scripts/ingest_open_data.py`. Each record includes district, building type, area (ping), monthly rent, transaction date, and optional floor/age fields. Production would add scheduled pulls from Taiwan government open-data APIs and licensed partnerships with listing platforms. Demand-validation inputs (surveys, forum codings) are stored as JSON and summarized by `scripts/collect_demand_evidence.py`.

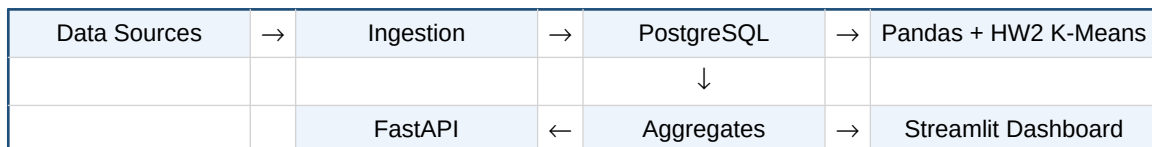
4.2 Storage and processing

Structured facts land in PostgreSQL (SQLite locally) via SQLAlchemy models. A batch processor (`pipeline/processor.py`) computes per-district medians and 25th/75th percentile rent-per-ping bands using Pandas. We also reuse Homework 2's MapReduce K-Means implementation (`pipeline/mapreduce_kmeans.py`, adapted from `hw2/hw_r14921059/mapper.py` and `reducer.py`) to segment the market into budget/value/premium/luxury clusters on features (`area_ping`, `rent_per_ping`). At 100× scale, Pandas and K-Means map to Spark batch jobs over parquet files in object storage.

4.3 Delivery

Customers consume the product through: (1) a Streamlit dashboard for interactive quotes, district charts, and K-Means segments; (2) a FastAPI REST layer with `/quote`, `/metrics/districts`, `/clusters`, and admin reprocess endpoints; (3) future Pro features — CSV export, Line/email alerts — backed by Redis pub/sub in the Docker Compose stack.

4.4 Architecture diagram



Layer	Technology	Rationale
Ingestion	Python, CSV/API adapters	Simple, testable, course-aligned
Storage	PostgreSQL / SQLite	Relational aggregates, indexed lookups
Processing	Pandas + HW2 MapReduce K-Means	District stats + market segments
API	FastAPI + Uvicorn	Typed REST delivery, easy deployment
UI	Streamlit + Plotly	Rapid SMB dashboard, demo-friendly
Ops	Docker Compose	Reproducible local and cloud deploy

4.5 Scalability and unit economics (optional)

At MVP (~500 transactions, 50 paying users), estimated infra cost is under USD 25/month on a small VM plus managed Postgres. Gross margin at $\text{NT\$}499 \times 50 \approx \text{NT\$}25\text{k/month}$ exceeds infra by an order of magnitude. At 100× data volume, batch processing shifts to Spark; API read load is served from precomputed aggregates, keeping query latency stable.

5. Go-to-Market Difficulties (Bonus Component 3)

- **Trust and adoption:** Landlords may trust agent relationships over an unknown SaaS dashboard. Mitigation: show transaction sample sizes, confidence scores, and side-by-side comparisons with their current manual method.
- **Data acquisition cost:** Government open data is free but delayed; listing platforms are fresher but license-bound. Over-reliance on scraped listings creates ToS and PDPA exposure.
- **Legal / privacy:** Even aggregate rental products must avoid exposing identifiable tenant or landlord records. Production requires PDPA review and clear retention policies.
- **Cold start:** Sparse districts yield wide bands and low confidence — early users in Neihu vs. Da'an may get uneven value. We surface confidence explicitly to set expectations.
- **Competition:** 591, HousePlus, or banks could bundle similar analytics. Our moat is narrow focus on micro-landlords, faster self-serve UX, and pricing aligned to their willingness to pay — not enterprise sales cycles.
- **Unit economics:** CAC via Facebook landlord groups is low, but churn rises if vacancies are rare (users pause subscriptions). Annual plans and vacancy-triggered alerts improve retention.

6. Business Model Summary

Free tier: read-only district snapshot. Pro (NT\$499/mo): custom quotes, exports, email support. API (NT\$999/mo): programmatic access for micro property managers. Optional one-off vacancy report (NT\$299) acts as conversion hook. Primary acquisition: PTT/LINE landlord communities, NTU alumni networks, and agent referral rebates.

7. Repository and Reproducibility

The GitHub repository (https://github.com/YOUR_USERNAME/leasepulse-taipei) includes ingestion scripts, batch pipeline, API, dashboard, sample data, Docker configuration, and demand-evidence reproduction steps in README.md. Run `python report/generate_report.py` to regenerate this PDF after updating evidence files.

8. Conclusion

LeasePulse Taipei demonstrates that course-scale big-data tooling — structured ingestion, SQL storage, batch analytics, and API/dashboard delivery — can support a credible data monetization story when paired with rigorous customer definition and demand evidence. The defensible claim is not 'big data magic,' but a focused product that saves measurable landlord research time and reduces costly vacancy mistakes.